SS-P-385a
January 31, 1964

SUPERSEDING
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FEDERAL SPECIFICATION

PIPE, STEEL, (CEMENT-MORTAR LINING AND REINFORCED CEMENT-MORTAR COATING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cement-mortar lined and reinforced cement-mortar coated welded steel pressure pipe with bell and spigot joints, from 4 to 42 inches in diameter for use in water transmission, distribution, and feeder pipelines.

1.2 Classification.

1.2.1 *Classes.* Pipe shall be furnished in one of the following classes as specified (6.2), or as modified by provisions of 6.3.

Class	Water working pressure (maximum)
	p.s.i.
100	100
150	150
200	200
250	250
300	300
350	350
400	400

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-W-418—Wire, Steel, Cold Drawn for Concrete Reinforcement.
RR-W-375 — Wire, Fabric, Steel, Welded (for Reinforced Concrete).
SS-A-281—Aggregate; (for) Portland-Cement-Concrete.
SS-C-192—Cement, Portland.
TT-P-86 — Paint: Red-Lead-Base, Ready-Mixed.
TT-P-636—Primer Coating, Synthetic, Wood and Ferrous Metal.
TT-P-645—Primer, Paint, Zinc-Chromate, Alkyd Type.
TT-P-664—Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issues in effect on date of invitation for bids shall apply.

American Society For Testing And Materials:

Designations:

A 139—Specifications for Electric-Fusion (Arc) Welded Steel Pipe.

A 245—Specifications for Flat Rolled Carbon Steel Sheets of Structural Quality.

A 283—Specification for Low and Intermediate Tensile Strength Carbon Steel Plates of Structural Quality (Plates 2 Inches and Under in Thickness).

C 39—Method of Test of Compressive Strength of Molded Concrete Cylinders.

C 156—Water Retention Efficiency of Liquid Membrane-Forming Compounds and Impermeable Sheet Materials for Curing Concrete.

C 309—Specifications for Liquid Membrane Forming Compounds for Curing Concrete.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

American Association of Railroads:

General Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago 5, Illinois.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington D. C. 20005.)

3. REQUIREMENTS

3.1 Component parts. The pipe shall consist of the following component parts: A welded sheet-steel or plate-steel cylinder with joints formed integrally with the steel cylinder or with steel joint rings welded to the ends; a dense cement-mortar lining; a dense, concentric, steel-reinforced exterior mortar coating; a self-centering bell and spigot joint with a circular preformed elastomeric gasket, so designed that the joint will be watertight under all conditions of service.

3.2 Material.

3.2.1 *Cement.* Unless otherwise specified, the cement used in mortar for lining and coating shall be type I or type II conforming to SS-C-192, (6.2 and 6.4).

3.2.2 *Aggregate.* The aggregate shall be in accordance with SS-A-281, class 1; except that for lining less than 1/2 inch in thickness, 100 percent of the aggregate shall pass the No. 4 sieve.

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3.2.3 *Steel.*

3.2.3.1 *Cylinders.* Steel for cylinders shall be hot-rolled, low carbon open hearth or electric furnace steel sheets conforming to ASTM Designation A 245, grade B or C; steel plates conforming to ASTM Designation A 283, either grade B or C; or steel pipe conforming to ASTM Designation A 139, grade A or B.

3.2.3.2 *Joint rings.* When welded joint rings are used, the material for bells and spigots may be the same as the steel used in the cylinder; or, the material may be rolled mill sections of AISI Grade Designation C 1012 with the following percentages of chemical composition:

Carbon	Manganese	Phosphorus	Sulfur
0.10-0.15	0.30-0.60	0.040 max.	0.050 max.

3.2.3.3 *Reinforcement.* Wire for helically wound or ribbon mesh reinforcement for the mortar coating shall conform to QQ-W-418 or RR-W-375 when welded wire fabric is used. Welded wire fabric shall be self-furring.

3.2.4 *Mortar.* The cement mortar used for the lining and coating shall be composed of mixtures of portland cement, aggregate and water, well mixed and of the proper consistency to produce dense, homogeneous lining and coating that will adhere firmly to the steel cylinder. The cement mortar shall develop compressive strengths of not less than 2,600 p.s.i. at 7 days and 4,500 p.s.i. at 28 days, when subjected to the test specified in 4.4.1.

3.2.4.1 *Water.* Water used for mixing mortar and for curing pipe shall be clean and free from injurious amounts of salts, silt, oil, acid, strong alkalies, vegetable matter, or other impurities.

3.2.4.2 *Mix proportions.* Mortar for the pipe lining shall consist of part 1 cement to not more than 3 parts of aggregate, by weight. Mortar for the pipe coating shall consist of 1 part cement to not more than 4 parts of aggregate, by weight. The proportions and the water-cement ratio shall be those which will give the best overall results with the particular material and method of placing used for the work.

3.2.5 *Elastomeric gasket.* The gaskets for joints shall be fabricated from an elastomeric compound, shall be circular, and shall be dense, homogeneous, and free from porosity, blisters, pitting, and other imperfections. The elastomeric compound shall consist of first grade natural rubber, synthetic rubber, or a suitable combination thereof. The gaskets shall be furnished with the pipe. Gaskets shall conform to the following physical requirements (4.4.4).

3.2.5.1 *Tensile strength.* The tensile strength of the compound shall be not less than 2,300 p.s.i. (4.4.4).

3.2.5.2 *Elongation at rupture.* Elongation at rupture shall be not less than 425 percent (4.4.4).

3.2.5.3 *Change in weight, liquid immersion.* Change in weight due to immersion in water for 2 days at 70°C. shall not exceed 5 percent (4.4.4).

3.2.5.4 *Compression set test.* The compression set shall not exceed 20 percent (4.4.4).

3.2.5.5 *Tensile strength after aging.* The tensile strength of the compound shall be not less than 80 percent of the tensile strength before aging after being subjected to an accelerated aging test in an air oven for 96 hours at 158° F., or in an oxygen bomb test for 48 hours, 158°F., 300 p.s.i. (4.4.4).

3.2.5.6 *Hardness.* The hardness of the compound shall be not less than 40 or more than 65 using the Shore A Durometer. The increase over original Shore durometer after the accelerated aging test (3.2.5.5 and 4.4.4) shall not exceed 8 (4.4.4).

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3.2.5.7 *Acetone extract.* Acetone extract shall be not more than 15 percent (4.4.4).

3.2.6 *Specials and bends.* The steel sheets and plates used in the fabrication of specials and bends shall conform to the requirements of 3.2.3.1. Steel reinforcement shall conform to the requirements of 3.2.3.3.

3.3 Design and fabrication.

3.3.1 *Steel cylinders.* Cylinders shall be formed by shaping and butt welding steel sheets or plates, or may be manual or submerged arc-welded straight-seam or spiral-seam steel pipe. The wall thicknesses shall be not less than those shown in table I for the respective nominal diameters and pressures except as provided in 6.3. Unless otherwise specified (6.2), the wall thickness of the steel cylinder shall be computed on the basis of having a maximum stress of 15,000 p.s.i. in the steel at the design pressure but shall be not less than the thickness of No. 14 gage. Joint ends shall be completed before hydrostatic testing, or if the cylinder has been hydrostatically tested before completion of the joint ends, an approved length of each joint end, except for end areas unaffected by internal working pressure, shall be given an equivalent hydrostatic test. All welding shall be done by an approved helical or straight seam process. Welds shall have complete penetration. Steel cylinders shall be hydrostatically tested (4.4.2) and the base metal shall be subject to tension tests as provided in 4.4.3.

3.3.2 *Exterior reinforcement.* The steel reinforcement in the mortar coating shall consist of: (a) Helically wound cold-drawn steel wire in accordance with QQ-W-418; or (b) A cage of self-furring welded steel wire fabric of 2- by 2- inch No. 14 gage fabric or 2- by 4- inch No. 13 gage fabric in accordance with RR-W-375; or (c) 1- inch hexagonal No. 18 gage or 1-1/2 inch hexagonal No. 17 gage ribbon mesh, twisted wire fabric, selvaged both edges using steel wire conforming to QQ-W-418. Helically wound wire shall be not less than the thickness of No. 14 gage, and shall be wound at maximum spacing of 1-1/4 inches. The reinforcement shall be embedded at the approximate center of the cement-mortar coating.

3.3.3 *Joint forming.* Bell and spigot ends shall be circular in shape and shall be so designed that the gasket will be restrained or confined to an annular space in such manner that movement of the pipe or hydrostatic pressure cannot displace the gasket. Compression of the gasket in joint closure position shall not be dependent upon water pressure in the pipe and the compressed gasket shall substantially fill the annular space and effect a watertight seal. The gasket shall be of such size that when the outer surface of the spigot and the inner surface of the bell come into contact at some portion of their peripheries the deformation in the gasket shall not exceed 50 percent at the point of contact and at the diametrically opposite point shall not be less than 15 percent of the stretched gasket diameter. The gasket shall be the sole element of the joint dependent upon to provide watertightness. The difference in circumferential measurement between the outside circumference of the spigot and the inside circumference of the bell shall be 0.060-inch minimum and 0.200-inch maximum as measured at the edge of a compressed gasket at nominal joint closure position. The unlined interior surface of the bell and the uncoated exterior surface of the spigot shall be protected by a corrosion-resistant paint in accordance with TT-P-86, TT-P-636, TT-P-645, or TT-P-664. Each end of joints formed integrally with the steel cylinder shall be formed either by sizing with a machined swage or die, or by rolling with suitable rolling equipment. Welded steel joint rings shall be formed by one or more pieces of steel welded together, either by resistance welding or by manual arc welding. After welding, the rings shall be smoothed by grinding the weld flush with the adjacent

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surface. Welded joint rings made or rolled mill sections shall be cold-expanded so that they are sized to the design diameter and within the tolerances specified herein (3.4.1).

3.3.4 *Specials and bends.* The manufacturer shall furnish specially fabricated specials and bends for closures, curves, bends and connections to valves or to other pipe. The specials and bends shall be fabricated from steel plate or sheets with cement-mortar lining and cement-mortar coating reinforced with welded steel wire fabric. Specials and bends shall be made from tested cylinders.

3.3.4.1 *Long radius curves.* Long radius curves, horizontal and vertical, may be made with welded mitered bends or by taking small angular deflections at the bell and spigot joints and using short length pipe sections. If deflections at the bell and spigot joints are used the manufacturer shall furnish a joint that will safely provide for unsymmetrical closure of spigot into bell so that there may be a 3/4-inch pull from a normal 1/4-inch mortar space on one side of the joint in pipe sizes through 24 inches, and so that in sizes larger than 24 inches, a total 1 inch joint deflection shall be permitted by reducing the normal 1/2 inch inside mortar space to 1/4 inch and pulling the opposite side of the pipe 3/4 inch from normal closure. However, the angular deflection at bell and spigot joints shall in no case exceed 5 degrees.

TABLE I—*Dimensional requirements*

Nominal inside diameter of lined pipe (inches)	Lining thickness (inches)[2]	Coating thickness (inches, minimum)	Nominal steel clyinder thickness—inches[1]						
			Class 100	Class 150	Class 200	Class 250	Class 300	Class 350	Class 400
4	5/16	1/2	.0747	.0747	.0747	.0747	.0747	.0747	.0747
6	5/16	1/2	.0747	.0747	.0747	.0747	.0747	.1046	.1046
8	5/16	1/2	.0747	.0747	.0747	.0747	.1046	.1046	.1345
10	5/16	1/2	.0747	.0747	.0747	.1046	.1345	.1345	.1875
12	5/16	1/2	.0747	.0747	.1046	.1345	.1345	.1875	.1875
14	3/8	5/8	.0747	.1046	.1046	.1345	.1875	.1875	.2500
16	3/8	5/8	.0747	.1046	.1345	.1875	.1875	.2500	.2500
18	3/8	5/8	.0747	.1046	.1345	.1875	.2500	.2500	.3125
20	1/2	3/4	.0747	.1345	.1875	.1875	.2500	.3125	.3125
21	1/2	3/4	.0747	.1345	.1875	.1875	.2500	.3125	.3125
22	1/2	3/4	.1046	.1345	.1875	.2500	.2500	.3125	.3750
24	1/2	3/4	.1046	.1345	.1875	.2500	.3125	.3125	.3750
26	1/2	3/4	.1046	.1875	.1875	.2500	.3125	.3750	.3750
27	1/2	3/4	.1046	.1875	.2500	.2500	.3125	.3750	.4375
28	1/2	3/4	.1046	.1875	.2500	.2500	.3125	.3750	.4375
30	1/2	3/4	.1046	.1875	.2500	.3125	.3750	.3750	.4375
32	1/2	3/4	.1345	.1875	.2500	.3125	.3750	.4375	.5000
33	1/2	3/4	.1345	.1875	.2500	.3125	.3750	.4375	.5000
34	1/2	3/4	.1345	.1875	.2500	.3125	.3750	.4375	.5000
36	1/2	3/4	.1345	.1875	.2500	.3750	.4375	.5000	.5625
38	1/2	3/4	.1345	.2500	.3125	.3750	.4375	.5000	.5625
39	1/2	3/4	.1875	.2500	.3125	.3750	.4375	.5000	.5625
40	1/2	3/4	.1875	.2500	.3125	.3750	.4375	.5000	.6250
42	1/2	3/4	.1875	.2500	.3125	.3750	.5000	.5625	.6250

[1] The minimum thickness of steel cylinders shall be not less than No. 14 gage (0.0747 inch).
[2] See 3.4.1 and 3.5.2 for tolerances.

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3.4 Dimensions. Unless otherwise specified (6.2), the pipe shall have standard nominal laying lengths of not less than 20, nor more than 40 feet, and except where shorter lengths are required for fittings, curves, and closures. Dimensions shall be in accordance with table I.

3.4.1 *Tolerance.* A tolerance of minus 1/4 inch in the nominal inside diameter of the lined pipe as specified in table I is permissible. Tolerance for field joints shall conform to the requirements of 3.3.3.

3.5 Lining and coating.

3.5.1 *Preparation of steel surfaces.* Before lining or coating is applied the cylinder shall be clean of heavy flaky rust, loose mill scale, dirt, grease, or other objectionable foreign substances.

3.5.2 *Cement-mortar lining.* The cement-mortar lining shall be applied by the centrifugal method or by a method obtaining equivalent results. When necessary to insure attainment of a circular pipe within the tolerance specified in 3.4.1, external stiffener rings shall be placed around the cylinder during the spinning of the lining. The mortar lining shall conform to 3.2.4.2 and shall be placed within the cylinder to the proper thickness as shown in table I. A tolerance of minus 1/16 inch will be permitted in the lining thickness. Lining over the gasket groove shall be not less than 3/16 inch. The process used in the application of the lining shall produce a smooth, dense, durable surface, free from sand pockets, voids, oversanded areas, blisters, and excessively cracked areas.

3.5.3 *Reinforced cement-mortar coating.* The cement-mortar coating shall be applied by pneumatic placement, extrusion, or by a method producing equivalent results. The thickness of the mortar coating shall be not less than that shown in table I. The mortar shall conform to 3.2.4. All reinforcement shall be covered with a minimum thickness of 3/16 inch of cement mortar.

3.6 Curing. The pipe shall be cured by water curing, steam curing, or a combination of both, or the coating may be cured by covering with a plastic membrane. Water curing and steam curing may be used interchangeably on a time-ratio basis of 4 hours water curing to 1 hour steam curing.

3.6.1 *Steam curing.* Where steam curing is used, the pipe shall be kept in steam maintained at a temperature of 130° to 150° F. for the specified period. The pipe shall be brought to the curing temperature at a rate of not more than 30° F. per hour provided that in no case shall the temperature exceed 100° F. within 2 hours after the mortar is applied. Steam curing time shall be computed after the pipe has reached the curing temperature. If the pipe is to be steam cured, each section of pipe shall be kept continually moist during the steam curing period. After steam curing, the pipe shall be protected from rapid drops in temperature which may damage the lining or coating.

3.6.2 *Water curing.* Where water curing is used the pipe shall be kept continually moist by spraying or other means for 7 days before being moved to the trenchside. The pipe shall not be allowed to dry either on the inside or outside surfaces during the curing period.

3.6.3 *Membrane curing.* Membrane curing of coatings shall be by wrapping the cement-mortar coating with clear or light-colored plastic sheeting immediately after the coating is applied. The moisture loss of the sheeting shall be not more than 20 grams per square foot per 7 days when tested in accordance with ASTM C 156. The sheeting shall be wrapped to provide a tight overlap of not less than 1 inch and the total wrapping shall be free of pinholes, cuts or tears to provide an effective seal against the escape of moisture from the mortar. The overlap shall be such that the minimum thick-

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ness of mortar coating shown in table I shall be maintained. The pipe shall be carefully moved to the storage area and any damage to the sheeting shall be promptly repaired to maintain an effective seal for at least 7 days.

3.6.4 *Pipe lining.* When the lining is to be water cured, the pipe ends shall be tightly capped with a waterproof cover immediately following completion of the lining of each section to prevent the escape of moisture from the interior of the pipe. When additional moisture is required to maintain a moist condition, water shall be introduced inside the pipe after the mortar has attained sufficient set so that the introduction of water will not damage the mortar. When steam curing is used, the pipe ends shall be tightly capped with a waterproof cover whenever steam curing is interrupted or completed. The waterproof covers shall be kept on the pipe throughout and beyond the curing period, the covers to become a component of the completed pipe section. The pipe lining shall be water cured for a minimum of 24 hours before application of the coating. Steam curing, when used, shall be on a time-ratio basis as provided in 3.6.

3.6.5 *Identification marking.* Each length of pipe, special and bend shall have plainly marked by painting, the nominal size, the class for which it is designed, the name of the manufacturer or his trademark, and the date of manufacture.

3.7 **Workmanship.** All pipe shall be uniform in construction and quality.

3.8 **Defective lining and coating.** Any defective area in the lining or coating shall be removed to the depth of defective mortar. Additional mortar shall be removed from defective coating as necessary to provide for keying of the repair mortar to the reinforcement. Defective areas shall be repaired by mechanical means or by hand application to the full required thickness. Reinforcement in the coating shall be repaired or replaced as necessary to assure a completely reinforced coating. The mortar used for repair of defective lining and coating shall have the same proportions of cement and sand as used in the lining or coating being repaired. Mortar repairs may be cured by prompt application of clear-or white-pigmented sealing compound conforming to ASTM C309, except at joint ends where the compound will interfere with the bond of joint mortar. Repaired pipe shall not be shipped or handled for a minimum of 24 hours following application of the sealing compound. Continued care shall be exercised thereafter to maintain the membrane seal. When there are numerous or large defective areas in a lining or coating, the entire lining or coating shall be removed and replaced.

3.9 **Protection.** During the entire period of the application of mortar lining and coating and the curing thereof, the mortar shall be protected from temperatures below 40° F., and the pipe sections shall be carefully supported and handled so as to avoid injury to the linings and coating. The pipe shall be properly stored to prevent damage to the linings and coatings, including excessive cracking or separation from the steel surfaces.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 **Sampling.** The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 *Lot.* Unless otherwise specified (6.2), a lot shall consist of all fabricated pipe of the same class and size offered for delivery at one time.

4.2.2 *Sampling for inspection of packing.* Packed pipe, ready for shipment, shall be sampled in accordance with level I.

4.2.3 *Sampling for dimensional and visual examination of pipe.* Sampling of pipe for these characteristics shall be in accordance with level L4.

4.2.4 *Sampling for tests.*

4.2.4.1 *Mortar.* Not more than four cylinders from each day's pour shall be taken for tests.

4.2.4.2 *Pipe.* Sampling of pipe for hydrostatic testing shall be in accordance with level L2.

4.2.4.3 *Tensile specimens.* When required, the number of specimens shall be one for every multiple of 3,000 feet of pipe or fraction thereof.

4.2.4.4 *Elastomeric gaskets.* Sampling of gaskets shall be in accordance with level L2.

4.3 **Inspection and tests.** The acceptable quality levels AQL's shall be in accordance with MIL-STD-105.

4.3.1 *Inspection.* Pipe selected in 4.2.2 and 4.2.3 shall be examined in accordance with table II. The AQL's for defects listed in table II shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE II—*Classification of defects*

Defect	Major	Minor
Spigot end—Not Corrosion resistant	X	
Spigot end—Dimension in error	X	
Bell end—Dimensions in error	X	
Inside diameter in error	X	
Inside lining thickness excessive	X	
Inside lining thickness below minimum	X	
Outside coating thickness below minimum	X	
Covers missing or not waterproof	X	
Mortar lining surface, rough, blistered sandpockets, voids, cracked or oversanded areas	X	
Separation of lining from steel surfaces	X	
Identification marking not complete		X
Identification marking missing, incorrect, or illegible	X	

4.3.2 *Lot acceptance tests.* Samples selected in 4.2.4 shall be subjected to the tests in 4.4, AQL 6.5 percent defective.

4.4 **Tests.**

4.4.1 *Compressive strength for mortar.* Test cylinders shall be cast samples of the mortar used in lining and coating the pipe. The curing of test cylinders shall be in conformity with the curing of the pipe. Cylinders shall be molded and tested in accordance with ASTM C 39 (3.2.4).

4.4.2 *Hydrostatic pressure.* Steel cylinders shall be subjected to a hydrostatic pressure test which stresses the steel to 22,000 p.s.i. While under pressure test, all welds shall be thoroughly inspected and all parts showing leakage shall be marked. Cylinders which show any leakage under test shall be rewelded at the points of leakage and subjected to another hydrostatic test (3.3.1).

4.4.3 *Base metal test for steel cylinders.* When specified (6.2), longitudinal tension test specimens shall be cut from the steel cylinders. The base metal test specimens shall be tested in accordance with method

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211.1 of Fed. Test Method Std. No. 151 to determine tensile strength. Cylinders from which test specimens are cut may be patch welded in an approved manner (3.3.1).

4.4.4 *Elastomeric gaskets.* The elastomeric gaskets shall be tested in accordance with table III.

TABLE III—*Test methods for elastomeric gaskets*

Fed. Test Method Std. No. 601	Title	Applicable requirement paragraph
Method		
3021	Hardness, durometer	3.2.5.6
3311	Compression set	3.2.5.4
4111	Tensile strength	3.2.5.1
4121	Elongation, ultimate	3.2.5.2
6251	Change in weight, liquid immersion	3.2.5.3
7111	Oxygen pressure test	3.2.5.5
7221	Air heat test	3.2.5.5
16321	Extract, acetone	3.2.5.7

4.4.5 *Specials and bends.* Specials and bends shall be subject to the test specified for the pipe (4.4.1 and 4.4.3), except that specials and bends made from pipe tested in accordance with 4.4.2 will not be subjected to further hydrostatic tests.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packing. Pipe shall be shipped loose in accordance with Section No. 2 of the association of American Railroads General Rules Governing the loading of Commodities on open top cars for rail shipment or in accordance with the National Motor Freight Classification or applicable exceptions thereto.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pipe covered by this specification is intended for use in water transmission, distribution and feeder pipelines for internal working pressures of 100 to 400 pounds per square inch. Thicknesses of the cylinder walls shown in table I are computed on the basis of an allowable fiber stress in the cylinder of 15,000 p.s.i. The thicknesses of the standard sheets and plates shown in the table are equal to or greater than the computed thicknesses. Local conditions may necessitate further design, installation, and operating considerations; such as, intermediate working pressures (6.3), transient pressure conditions, pipe bedding, depth of backfill, terrain, and prevailing soil conditions.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class and nominal inside diameter of pipe (1.2 and table I).
(*c*) Type of cement (3.2.1 and 6.4).
(*d*) Steel cylinder thickness and design stress if different from that prescribed (3.3.1, 6.1 and 6.3).
(*e*) Laying lengths of pipe if other than provided by 3.4.
(*f*) Size of lot, if different from 4.2.1.
(*g*) Whether tensile tests are required for the base metal (4.4.3).

6.3 Intermediate working pressures and steel cylinder thicknesses. Pipe for intermediate working pressures may be specified. Steel cylinders with thickness differ-

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ing from those indicated in table I may be furnished for the pressure ratings shown and for specified intermediate pressure ratings. The design stress in no case should exceed 15,000 p.s.i. and the minimum thickness should be not less than No. 14 gage.

6.4 Selecting type of cement. In many locations where mortar lined and coated steel pipe is to be used, particularly in the western part of the United States, soils and ground water that will be in contact with the mortar coating, and water that will be flowing through the pipe in contact with the mortar lining, contain water-soluble sulfates in sufficient quantity to cause rapid deterioration of the mortar from sulfate attack. Use of type I cement in SS-C-192 for mortar will not provide the necessary protection in such cases.

Type II (Moderate sulfate resisting) cement should be required when water soluble sulfates (as SO_4) in soil are in the range of 0.1 to 0.2 percent and, for water, in the range of 150 to 1,000 parts per million. Type V (sulfate resisting) cement should be used when soils contain in excess of 0.2 percent water-soluble sulfate and water samples contain in excess of 1,000 parts per million. Cement low in alkalies is necessary when the aggregates to be used in the linings and coating are sufficiently reactive (reaction between alkalies in cement and reactive type aggregate) to cause disruptive expansion of the hardened mortar. SS-C-192 contains requirements for low alkali cement.

6.5 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Pipe, steel, mortar lined and coated. Carload minimum weight 40,000 pounds.

Motor:

Pipe, steel, mortar covered and lined. Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—MO

Navy—Ships

Air Force—MOA

Copies of this specification may be purchased for 10 cents each.

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